ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2023 P2HR Worked Solutions

Jun 2023 | P2HR

33 questions ▪ question image followed by worked solution

Prepared by

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Student Edition

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 4

TOPIC: **Expanding Brackets**

Jun 2023 P2HR - Jun 2023 | P2HR

1 $P = m^2 - 4c$

(a) Work out the value of P when $m = -5$ and $c = 3$

$P =$ ....

(2)

(b) Expand and simplify $(x + 5)(x - 7)$

....

(2)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Expanding Brackets**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Expanding brackets. The tag is correct.

Solution

$$P = m^2 - 4c = (-5)^2 - 4(3) = 25 - 12 = 13$$

$$(x+5)(x-7) = x^2 - 7x + 5x - 35 = x^2 - 2x - 35$$

FINAL ANSWER

$P = 13$, and $x^2 - 2x - 35$

PAST PAPER QUESTION

Q#: 2 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

Jun 2023 P2HR - Jun 2023 | P2HR

2 Sandeep wants to buy some packets of pens and some boxes of pencils for his stationery shop.

Each packet of pens contains 9 pens.
Each box of pencils contains 12 pencils.

Each packet of pens costs £7.60
Each box of pencils costs £4.80

Sandeep can only buy full packets of pens and full boxes of pencils.
He wants to buy exactly the same number of pens as pencils.

Work out the minimum amount Sandeep needs to pay.

£..

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

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Topic check. Corrected to Prime Factors, HCF & LCM.

Solution

Pens come in packs of 9 and pencils come in boxes of 12.
The smallest equal number of pens and pencils is

$$\mathrm{LCM}(9, 12) = 36$$

So Sandeep buys

$$4 \text{ packs of pens}$$

and

$$3 \text{ boxes of pencils}$$

Cost:

$$4(7.60) + 3(4.80) = 30.40 + 14.40 = 44.80$$

FINAL ANSWER

£44.80.

PAST PAPER QUESTION

TOPIC: **Standard & Compound Units**

Jun 2023 P2HR - Jun 2023 | P2HR

Q#: 3 Marks: 3

3 Anjali travels on the Eurostar train from Paris to Amsterdam.

The distance the train travels between Paris and Amsterdam is 515 km.
The time taken by the train to travel between Paris and Amsterdam is 3 hours 18 minutes.

Work out the average speed of the train.
Give your answer in km/h correct to the nearest whole number.

.... km/h

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Standard & Compound Units**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

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Topic check. Corrected to Standard & Compound Units.

Solution

Time = 3 hours 18 minutes:

$$3 + \frac{18}{60} = 3.3 \text{ hours}$$

$$\text{average speed} = \frac{515}{3.3} = 156.06\ldots$$

FINAL ANSWER

156 km/h.

PAST PAPER QUESTION

Q#: 4 Marks: 2

TOPIC: **Sequences**

Jun 2023 P2HR - Jun 2023 | P2HR

4 Here are the first four terms of an arithmetic sequence.

38 31 24 17

Find an expression, in terms of n, for the nth term of the sequence.

....

(Total for Question 4 is 2 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 2

TOPIC: **Sequences**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

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Topic check. Sequences. The tag is correct.

Solution

The sequence is

$$38,\ 31,\ 24,\ 17$$

The common difference is -7.

$$\text{nth term} = 38 + (n-1)(-7)$$

$$= 38 - 7n + 7 = 45 - 7n$$

FINAL ANSWER

$45 - 7n$.

PAST PAPER QUESTION

TOPIC: **Area & Perimeter**

Jun 2023 P2HR - Jun 2023 | P2HR

Q#: 5 Marks: 4

5 A field is in the shape of a trapezium.

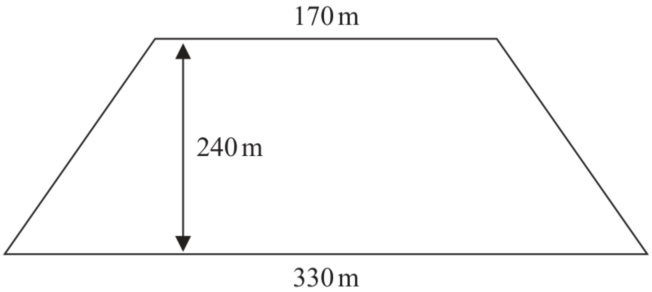

Diagram **NOT** accurately drawn

The field is sold for a price of \$49 650

Given that 1 hectare = 10 000 m^2

work out the average price of the field per hectare.

\$...

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Area & Perimeter**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

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Topic check. Area & Perimeter. The tag is correct.

Solution

$$\text{area} = \frac{1}{2}(170 + 330) \times 240 = 60000 \text{ m}^2$$

$$60000 \text{ m}^2 = 6 \text{ hectares}$$

$$\text{price per hectare} = 49650 \div 6 = 8275$$

FINAL ANSWER
\$8275.

PAST PAPER QUESTION

Q#: 6 Marks: 7

TOPIC: Percentages

Jun 2023 P2HR - Jun 2023 | P2HR

6 In his previous job, Pierre was paid 400 euros in total for working a 5-day week.

In his new job, Pierre is paid 14 euros per hour.
In his new job, Pierre works for 7 hours each day for a 5-day week.

(a) Work out the percentage increase in the amount that Pierre is paid for a 5-day week.

....%

(4)

Marie changes her job.

Her salary decreases by 6%
Her new salary is 23 030 euros.

(b) Work out Marie's salary before she changes her job.

.... euros

(3)

(Total for Question 6 is 7 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 7

TOPIC: **Percentages**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

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Topic check. Corrected to Percentages. The question is about percentage increase and reverse percentage decrease.

Method

Pierre's old weekly pay is 400 euros.
His new weekly pay is

$$14 \times 7 \times 5 = 490$$

The increase is

$$490 - 400 = 90$$

Percentage increase:

$$\frac{90}{400} \times 100 = 22.5\%$$

Marie's new salary is after a 6% decrease, so it is 94% of her old salary.

$$0.94 \times \text{old salary} = 23030$$

$$\text{old salary} = 23030 \div 0.94 = 24500$$

FINAL ANSWER

Pierre's pay increased by 22.5%. Marie's old salary was 24500 euros.

PAST PAPER QUESTION

Q#: 7 Marks: 2

TOPIC: **Algebraic Roots & Indices**

Jun 2023 P2HR - Jun 2023 | P2HR

7 (a) Simplify $(4^{-2})^0$

....

(1)

$3^{-14} \times 3^8 = 3^m$

(b) Find the value of m

$m =$ ....

(1)

(Total for Question 7 is 2 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 2

TOPIC: **Algebraic Roots & Indices**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

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Topic check. Algebraic Roots and Indices. The tag is correct.

Method

Any non-zero expression to the power 0 is 1:

$$(4^{-2})^0 = 1$$

For part (b),

$$3^{-14} \times 3^8 = 3^{-14+8} = 3^{-6}$$

So

$$m = -6$$

FINAL ANSWER

(a) 1, (b) $m = -6$.

PAST PAPER QUESTION

Topic: **Graphing Inequalities**

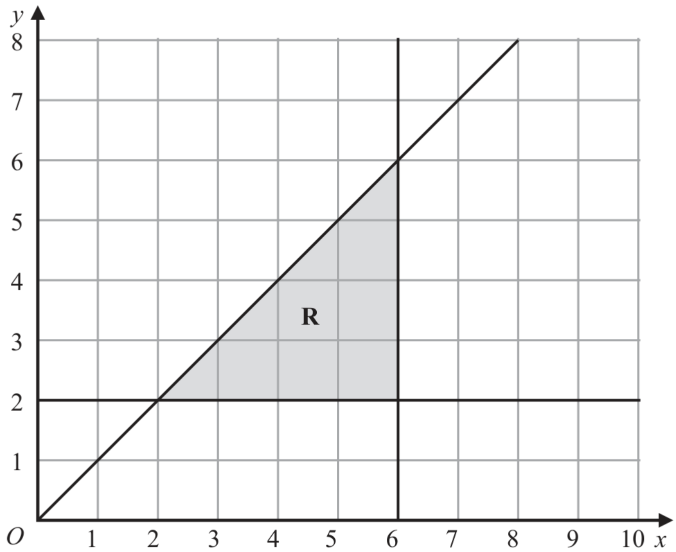
Jun 2023 P2HR - Jun 2023 | P2HR

Q#: 8 Marks: 5

8 (a) Solve $9 - 4x > 17$

....

(2)



(b) Write down the three inequalities that represent the shaded region **R**

....

....

....

(3)

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 5

TOPIC: **Graphing Inequalities**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct. This combines solving and graphing inequalities.

Solution

For part (a),

$$9 - 4x > 17$$

$$-4x > 8$$

$$x < -2$$

For part (b), the region is above $y = 2$, left of $x = 6$, and below $y = x$:

$$y \geq 2$$

$$x \leq 6$$

$$y \leq x$$

FINAL ANSWER

(a) $x < -2$. (b) $y \geq 2$, $x \leq 6$, $y \leq x$.

PAST PAPER QUESTION

Q#: 9 Marks: 6

Topic: **Volume & Surface Area**

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9 The diagram shows a rectangular sheet of metal *ABCD*

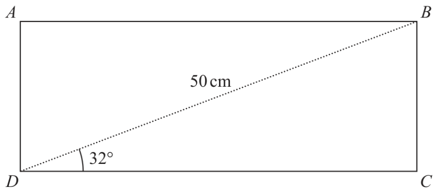

Diagram **NOT** accurately drawn

$BD = 50$ cm and angle $BDC = 32°$

Nasser joins side *AD* to side *BC* to form a cylinder.

BC is the height of the cylinder.
DC is the circumference of the cross section of the cylinder.

Work out the volume, in cm^3, of the cylinder.
Give your answer correct to 3 significant figures.

.... cm^3

(Total for Question 9 is 6 marks)

PAST PAPER SOLUTION

Q#: 9 **Marks: 6**

TOPIC: **Volume & Surface Area**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

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Topic check. Corrected to Volume & Surface Area.

Solution

In triangle DBC,

$$DC = 50\cos 32^\circ = 42.402\ldots, \qquad BC = 50\sin 32^\circ = 26.496\ldots$$

The circumference of the cylinder is DC, so

$$2\pi r = DC$$

$$r = \frac{42.402\ldots}{2\pi}$$

Volume:

$$V = \pi r^2 h = \pi\left(\frac{42.402\ldots}{2\pi}\right)^2 (26.496\ldots) = 3790.974\ldots$$

FINAL ANSWER

3.79×10^3 cm^3 to 3 significant figures.

PAST PAPER QUESTION

Q#: 10 Marks: 3

TOPIC: **Statistics Toolkit**

Jun 2023 P2HR - Jun 2023 | P2HR

10 Gemara works as a taxi driver.
Last week, he recorded the following information about the distances he drove.

For the 5 days from Monday to Friday, the mean number of kilometres he drove was 104
For the 7 days from Monday to Sunday, the mean number of kilometres he drove was 127

On Saturday, Gemara drove 132 kilometres.

Work out the number of kilometres he drove on Sunday.

.... kilometres

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Statistics Toolkit**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

Total for Monday to Friday:

$$5 \times 104 = 520$$

Total for Monday to Sunday:

$$7 \times 127 = 889$$

So the total for Saturday and Sunday is

$$889 - 520 = 369$$

Saturday was 132 km, so

$$369 - 132 = 237$$

FINAL ANSWER

237 km.

PAST PAPER QUESTION

TOPIC: **Algebraic Roots & Indices**

Jun 2023 P2HR - Jun 2023 | P2HR

Q#: 11 Marks: 2

11 Express $\left(\dfrac{m^6k^{10}}{25}\right)^{\frac{3}{2}}$ in the form $\dfrac{m^ak^b}{c}$ where a, b and c are integers to be found.

....

(Total for Question 11 is 2 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 2

TOPIC: **Algebraic Roots & Indices**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to Algebraic Roots & Indices. This is an algebraic fractional-index question, not a standard form question.

Method

$$\left(\frac{m^6k^{10}}{25}\right)^{\frac{3}{2}} = \frac{(m^6)^{\frac{3}{2}}(k^{10})^{\frac{3}{2}}}{25^{\frac{3}{2}}}$$

$$(m^6)^{\frac{3}{2}} = m^9$$

$$(k^{10})^{\frac{3}{2}} = k^{15}$$

Also,

$$25^{\frac{3}{2}} = (\sqrt{25})^3 = 5^3 = 125$$

So

$$\left(\frac{m^6k^{10}}{25}\right)^{\frac{3}{2}} = \frac{m^9k^{15}}{125}$$

FINAL ANSWER

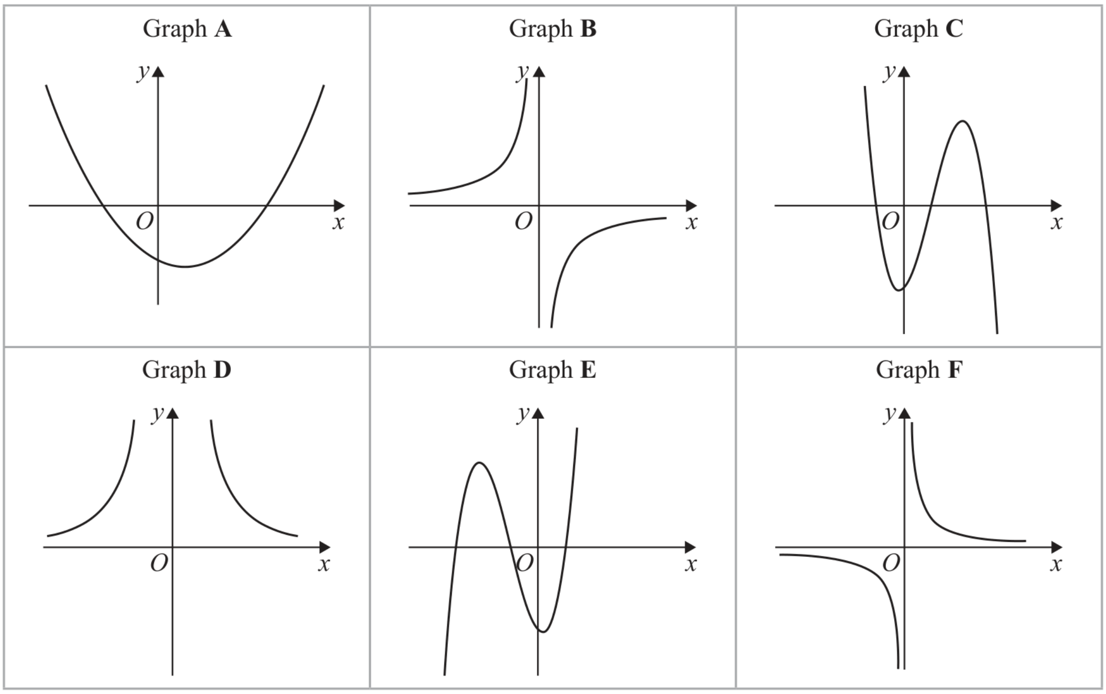
$$\frac{m^9k^{15}}{125}$$

PAST PAPER QUESTION Q#: 12 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2023 P2HR - Jun 2023 | P2HR

12 Here are six graphs.



Write down the letter of the graph of

(a) $y = \dfrac{10}{x^2}$

....

(1)

(b) $y = x - 3 + 3x^2 - x^3$

....

(1)

(c) $y = -\dfrac{3}{x}$

....

(1)

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

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Topic check. Corrected to graphs of functions. This is matching equations to graph shapes.

Solution

For

$$y = \frac{10}{x^2}$$

the graph is positive on both sides and symmetrical in the y-axis.
This is Graph D.
For

$$y = x - 3 + 3x^2 - x^3$$

the leading term is $-x^3$, so it is a cubic falling to the right with two turning points.
This is Graph C.
For

$$y = -\frac{3}{x}$$

the graph is a negative reciprocal graph in quadrants 2 and 4.
This is Graph B.

FINAL ANSWER

D, C, B.

PAST PAPER QUESTION

Q#: 13 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2023 P2HR - Jun 2023 | P2HR

13 Feruzi invests 80 000 Kenyan shillings (KES)
He invests the money for 3 years at x% compound interest each year.

At the end of 3 years, the total interest he receives is 6151.25 KES

Work out the value of x

$x =$ ....

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

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Topic check. Compound Interest and Depreciation. The tag is correct.

Method

The final value is

$$80000 + 6151.25 = 86151.25$$

Let the annual compound interest rate be $x\%$.

$$80000\left(1+\frac{x}{100}\right)^3 = 86151.25$$

$$1+\frac{x}{100} = \sqrt[3]{\frac{86151.25}{80000}} = 1.025$$

$$x = 2.5$$

FINAL ANSWER

$x = 2.5$

PAST PAPER QUESTION Q#: 14 Marks: 4

TOPIC: **Statistics Toolkit**

Jun 2023 P2HR - Jun 2023 | P2HR

14 Akari played a computer game eleven times.
Here are her scores.

25 20 28 27 26 22 23 29 20 29 26

(a) Find the interquartile range of her scores.

....

(3)

Machi played the same computer game eleven times.

The interquartile range for Machi's scores was 9

(b) Who had the more consistent scores, Akari or Machi?
Give a reason for your answer.

....

....

(1)

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 4

TOPIC: **Statistics Toolkit**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Statistics Toolkit. The tag is correct.

Solution

Akari's scores in order are

$$20,\ 20,\ 22,\ 23,\ 25,\ 26,\ 26,\ 27,\ 28,\ 29,\ 29$$

$$Q_1 = 22, \qquad Q_3 = 28$$

$$\text{IQR} = 28 - 22 = 6$$

Machi's interquartile range is 9.
Akari is more consistent because her interquartile range is smaller.

FINAL ANSWER

Akari's IQR is 6, and Akari is more consistent.

PAST PAPER QUESTION

Q#: 15 **Marks: 3**

TOPIC: **Probability Toolkit**

Jun 2023 P2HR - Jun 2023 | P2HR

15 Osvaldo has a biased coin.
He spins the coin three times.

The probability that the coin lands on a head three times is $\frac{27}{64}$

Work out the probability that the coin will land on a tail three times.

....

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Toolkit. The tag is correct.

Solution

$$P(\text{three heads}) = \frac{27}{64}$$

So

$$P(H)^3 = \frac{27}{64}$$

$$P(H) = \frac{3}{4}$$

Therefore

$$P(T) = \frac{1}{4}$$

$$P(\text{three tails}) = \left(\frac{1}{4}\right)^3 = \frac{1}{64}$$

FINAL ANSWER

$\frac{1}{64}$.

PAST PAPER QUESTION

Topic: **Algebraic Proof**

Jun 2023 P2HR - Jun 2023 | P2HR

Q#: 16 Marks: 3

16 Show that $\dfrac{2\sqrt{3}}{\sqrt{3}-1}$ can be written in the form $a + \sqrt{a}$ where a is an integer.

Show your working clearly.

(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 3

TOPIC: **Algebraic Proof**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic proof. The tag is acceptable because the question asks for exact-form proof.

Solution

$$\frac{2\sqrt{3}}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{2\sqrt{3}(\sqrt{3}+1)}{3-1}$$

$$= \frac{6+2\sqrt{3}}{2} = 3+\sqrt{3}$$

This is in the form $a+\sqrt{a}$, where $a=3$.

FINAL ANSWER

$3+\sqrt{3}$, so $a=3$.

PAST PAPER QUESTION Q#: 17 Marks: 4

TOPIC: **Rearranging Formulas**

Jun 2023 P2HR - Jun 2023 | P2HR

17 Make x the subject of $y = \sqrt[3]{\dfrac{6 + 5x}{x + 4}}$

....

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 4

TOPIC: **Rearranging Formulas**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Rearranging formulas. The tag is correct.

Solution

$$y = \sqrt[3]{\frac{6+5x}{x+4}}$$

$$y^3 = \frac{6+5x}{x+4}$$

$$y^3(x+4) = 6+5x$$

$$x(y^3-5) = 6-4y^3$$

FINAL ANSWER

$x = \frac{6-4y^3}{y^3-5}$

PAST PAPER QUESTION Q#: 18 Marks: 3

Topic: **Circle Theorems**

Jun 2023 P2HR - Jun 2023 | P2HR

18

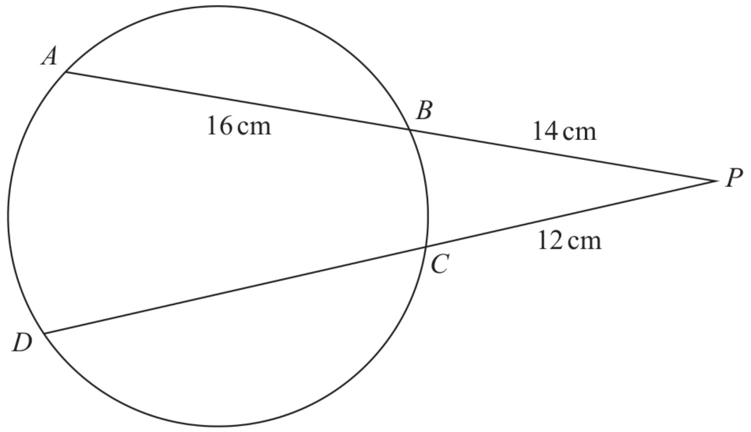

Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.

ABP and DCP are straight lines.

$AB = 16\,\text{cm}$ $\quad BP = 14\,\text{cm}$ $\quad CP = 12\,\text{cm}$

Work out the length of DC

.... cm

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 3

TOPIC: **Circle Theorems**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

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Topic check. Corrected to circle theorems. The question uses the intersecting secants theorem.

Solution

From the external point P, two secants meet the circle.
For the upper secant,

$$PB = 14, \qquad PA = 14 + 16 = 30$$

For the lower secant,

$$PC = 12, \qquad PD = 12 + DC$$

Using the secant theorem:

$$PB \times PA = PC \times PD$$

$$14 \times 30 = 12(12 + DC)$$

$$420 = 144 + 12DC$$

$$276 = 12DC$$

$$DC = 23$$

FINAL ANSWER
$DC = 23$ cm.

PAST PAPER QUESTION

Q#: 19 Marks: 4

TOPIC: **Probability Toolkit**

Jun 2023 P2HR - Jun 2023 | P2HR

19 30 adults booked to stay in a hotel.

19 adults booked breakfast
15 adults booked dinner
4 adults did not book breakfast or dinner

Some adults booked breakfast **and** dinner.

Meihui chooses at random two of the 30 adults.

Work out the probability that these two adults each booked breakfast **and** dinner.

....

(Total for Question 19 is 4 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 4

TOPIC: **Probability Toolkit**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Toolkit. The tag is correct.

Solution

There are 30 adults.

$$n(B) = 19, \quad n(D) = 15, \quad n(\text{neither}) = 4$$

So

$$n(B \cup D) = 30 - 4 = 26$$

$$n(B \cap D) = 19 + 15 - 26 = 8$$

The probability that two adults both had breakfast and dinner is

$$\frac{8}{30} \times \frac{7}{29} = \frac{28}{435}$$

FINAL ANSWER

$\frac{28}{435}$.

PAST PAPER QUESTION Q#: 20 Marks: 2

TOPIC: **Circle Theorems**

Jun 2023 P2HR - Jun 2023 | P2HR

20 *A*, *B* and *C* are points on a circle.

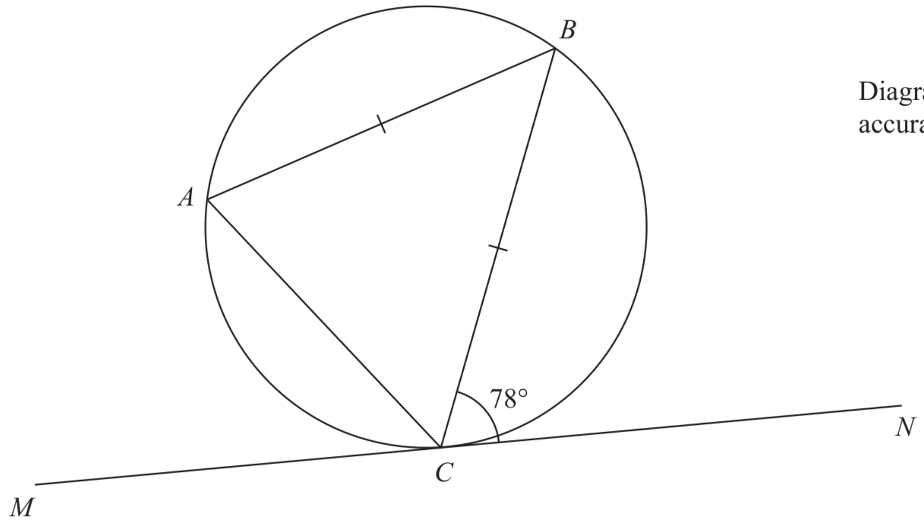

Diagram **NOT** accurately drawn

MN is the tangent to the circle at *C*

$AB = CB$

Angle $BCN = 78°$

Find the size of angle *ABC*

.... °

(Total for Question 20 is 2 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 2

TOPIC: **Circle Theorems**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The angle between the tangent CN and the chord CB is $78°$.
By the alternate segment theorem,

$$\angle BAC = 78°$$

Also,

$$AB = CB$$

so triangle ABC is isosceles and the base angles are equal:

$$\angle BAC = \angle ACB = 78°$$

Using the angle sum of a triangle:

$$\angle ABC = 180° - 78° - 78°$$

$$\angle ABC = 24°$$

FINAL ANSWER

$\boxed{24°}$

PAST PAPER QUESTION

TOPIC: **Circle Theorems**

Jun 2023 P2HR - Jun 2023 | P2HR

Q#: 20 **Marks: 2**

20 *A*, *B* and *C* are points on a circle.

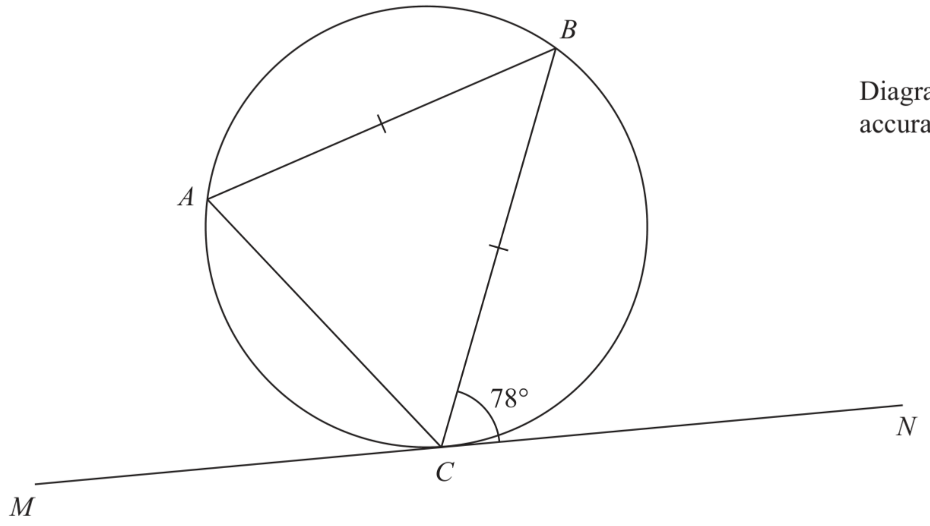

Diagram **NOT** accurately drawn

MN is the tangent to the circle at *C*

$AB = CB$

Angle $BCN = 78°$

Find the size of angle *ABC*

....°

(Total for Question 20 is 2 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 2

TOPIC: **Circle Theorems**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The angle between the tangent CN and the chord CB is $78°$.
By the alternate segment theorem,

$$\angle BAC = 78°$$

Also,

$$AB = CB$$

so triangle ABC is isosceles and the base angles are equal:

$$\angle BAC = \angle ACB = 78°$$

Using the angle sum of a triangle:

$$\angle ABC = 180° - 78° - 78°$$

$$\angle ABC = 24°$$

FINAL ANSWER

$\boxed{24°}$

PAST PAPER QUESTION

Q#: 21 **Marks: 5**

TOPIC: **Coordinate Geometry**

Jun 2023 P2HR - Jun 2023 | P2HR

21 Work out the coordinates of the points of intersection of

$$y - 2x = 1 \quad \text{and} \quad y^2 + xy = 7$$

Show clear algebraic working.

(.......................... , ...)

(.......................... , ...)

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION Q#: 21 Marks: 5

TOPIC: **Coordinate Geometry**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The line is

$$y - 2x = 1$$

So

$$y = 2x + 1$$

Substitute into

$$y^2 + xy = 7$$

$$(2x+1)^2 + x(2x+1) = 7$$

$$4x^2 + 4x + 1 + 2x^2 + x = 7$$

$$6x^2 + 5x - 6 = 0$$

Factorise:

$$(3x-2)(2x+3) = 0$$

So

$$x = \frac{2}{3} \quad \text{or} \quad x = -\frac{3}{2}$$

Using $y = 2x + 1$:

$$x = \frac{2}{3} \Rightarrow y = \frac{7}{3}$$

$$x = -\frac{3}{2} \Rightarrow y = -2$$

FINAL ANSWER

$\left(\frac{2}{3}, \frac{7}{3}\right)$ and $\left(-\frac{3}{2}, -2\right)$

PAST PAPER QUESTION

Q#: 21 **Marks: 5**

TOPIC: **Coordinate Geometry**

Jun 2023 P2HR - Jun 2023 | P2HR

21 Work out the coordinates of the points of intersection of

$$y - 2x = 1 \quad \text{and} \quad y^2 + xy = 7$$

Show clear algebraic working.

(.......................... , ...)

(.......................... , ...)

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION Q#: 21 Marks: 5

TOPIC: **Coordinate Geometry**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The line is

$$y - 2x = 1$$

So

$$y = 2x + 1$$

Substitute into

$$y^2 + xy = 7$$

$$(2x+1)^2 + x(2x+1) = 7$$

$$4x^2 + 4x + 1 + 2x^2 + x = 7$$

$$6x^2 + 5x - 6 = 0$$

Factorise:

$$(3x-2)(2x+3) = 0$$

So

$$x = \frac{2}{3} \quad \text{or} \quad x = -\frac{3}{2}$$

Using $y = 2x + 1$:

$$x = \frac{2}{3} \Rightarrow y = \frac{7}{3}$$

$$x = -\frac{3}{2} \Rightarrow y = -2$$

FINAL ANSWER

$\left(\frac{2}{3}, \frac{7}{3}\right)$ and $\left(-\frac{3}{2}, -2\right)$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2023 P2HR - Jun 2023 | P2HR

22 Here is a cuboid *ABCDEFGH*

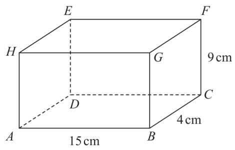

Diagram **NOT** accurately drawn

$AB = 15$ cm $\quad BC = 4$ cm $\quad CF = 9$ cm

(a) Work out the length of *BE*
Give your answer correct to 3 significant figures.

.... cm
(2)

Here is a cuboid *PQRSTUVW*

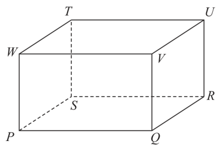

Diagram **NOT** accurately drawn

$PR = 42$ cm

The size of the angle between *PU* and the plane *PQRS* is 30°

M is the midpoint of *PR*

(b) Work out the size of angle *UMR*
Give your answer correct to 3 significant figures.

.... °
(3)

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

For part (a), BE is the space diagonal using the three perpendicular lengths:

$$AB = 15, \qquad BC = 4, \qquad CF = 9$$

$$BE^2 = 15^2 + 4^2 + 9^2$$

$$BE^2 = 225 + 16 + 81 = 322$$

$$BE = \sqrt{322} = 17.9 \text{ cm}$$

For part (b), the projection of PU onto the plane $PQRS$ is PR.

$$PR = 42$$

The angle between PU and the plane is 30°, so

$$\tan 30^\circ = \frac{RU}{PR}$$

$$RU = 42 \tan 30^\circ$$

Since M is the midpoint of PR,

$$MR = 21$$

In right triangle MRU,

$$\tan \angle UMR = \frac{RU}{MR}$$

$$\tan \angle UMR = \frac{42 \tan 30^\circ}{21}$$

$$\tan \angle UMR = 2 \tan 30^\circ$$

$$\angle UMR = 49.1^\circ$$

FINAL ANSWER

$BE = 17.9$ cm, $\angle UMR = 49.1^\circ$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2023 P2HR - Jun 2023 | P2HR

22 Here is a cuboid *ABCDEFGH*

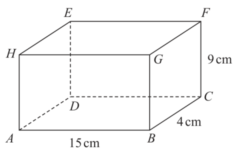

Diagram **NOT** accurately drawn

$AB = 15$ cm $\quad BC = 4$ cm $\quad CF = 9$ cm

(a) Work out the length of *BE*
Give your answer correct to 3 significant figures.

.... cm
(2)

Here is a cuboid *PQRSTUVW*

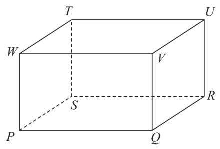

Diagram **NOT** accurately drawn

$PR = 42$ cm

The size of the angle between *PU* and the plane *PQRS* is 30°

M is the midpoint of *PR*

(b) Work out the size of angle *UMR*
Give your answer correct to 3 significant figures.

.... °
(3)

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

For part (a), BE is the space diagonal using the three perpendicular lengths:

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$$BE^2 = 225 + 16 + 81 = 322$$

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$$PR = 42$$

The angle between PU and the plane is $30°$, so

$$\tan 30° = \frac{RU}{PR}$$

$$RU = 42 \tan 30°$$

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$$MR = 21$$

In right triangle MRU,

$$\tan \angle UMR = \frac{RU}{MR}$$

$$\tan \angle UMR = \frac{42 \tan 30°}{21}$$

$$\tan \angle UMR = 2 \tan 30°$$

$$\angle UMR = 49.1°$$

FINAL ANSWER

$BE = 17.9$ cm, $\angle UMR = 49.1°$

PAST PAPER QUESTION

TOPIC: **Sequences**

Jun 2023 P2HR - Jun 2023 | P2HR

Q#: 23 Marks: 5

23 Here are the first three terms of an arithmetic sequence.

$$8p \qquad 7p - 3 \qquad 4p + 2$$

The sum of the first n terms of the sequence is -1914

Work out the value of n
Show your working clearly.

$n =$ ....

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Sequences**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The first three terms are

$$8p, \quad 7p-3, \quad 4p+2$$

For an arithmetic sequence, consecutive differences are equal:

$$(7p-3)-8p=(4p+2)-(7p-3)$$

$$-p-3=-3p+5$$

$$2p=8$$

$$p=4$$

So the first term is

$$8p=32$$

and the common difference is

$$(7p-3)-8p=25-32=-7$$

The sum of the first n terms is

$$S_n=\frac{n}{2}(2a+(n-1)d)$$

$$-1914=\frac{n}{2}(64-7(n-1))$$

$$-1914=\frac{n}{2}(71-7n)$$

$$-3828=n(71-7n)$$

$$7n^2-71n-3828=0$$

$$n=\frac{71\pm 335}{14}$$

Only the positive value is possible:

$$n=29$$

FINAL ANSWER
$n = 29$

PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Sequences**

Jun 2023 P2HR - Jun 2023 | P2HR

23 Here are the first three terms of an arithmetic sequence.

$$8p \qquad 7p - 3 \qquad 4p + 2$$

The sum of the first n terms of the sequence is -1914

Work out the value of n
Show your working clearly.

$n =$ ....

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Sequences**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The first three terms are

$$8p, \quad 7p - 3, \quad 4p + 2$$

For an arithmetic sequence, consecutive differences are equal:

$$(7p - 3) - 8p = (4p + 2) - (7p - 3)$$

$$-p - 3 = -3p + 5$$

$$2p = 8$$

$$p = 4$$

So the first term is

$$8p = 32$$

and the common difference is

$$(7p - 3) - 8p = 25 - 32 = -7$$

The sum of the first n terms is

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$-1914 = \frac{n}{2}(64 - 7(n-1))$$

$$-1914 = \frac{n}{2}(71 - 7n)$$

$$-3828 = n(71 - 7n)$$

$$7n^2 - 71n - 3828 = 0$$

$$n = \frac{71 \pm 335}{14}$$

Only the positive value is possible:

$$n = 29$$

FINAL ANSWER

$n = 29$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Volume & Surface Area**

Jun 2023 P2HR - Jun 2023 | P2HR

24 The surface area of sphere **A** is nine times the surface area of sphere **B**
The difference between the volume of sphere **A** and the volume of sphere **B** is 117π cm^3

Find the radius of the smaller sphere.
Show your working clearly.

.... cm

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION Q#: 24 Marks: 5

TOPIC: **Volume & Surface Area**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

Let the radius of the smaller sphere be r.
Let the radius of the larger sphere be R.
Surface area of a sphere is

$$4\pi r^2$$

The surface area of sphere A is nine times the surface area of sphere B, so

$$4\pi R^2 = 9(4\pi r^2)$$

$$R^2 = 9r^2$$

$$R = 3r$$

The difference between the volumes is 117π:

$$\frac{4}{3}\pi(3r)^3 - \frac{4}{3}\pi r^3 = 117\pi$$

$$\frac{4}{3}\pi(27r^3 - r^3) = 117\pi$$

$$\frac{104}{3}\pi r^3 = 117\pi$$

Cancel π:

$$\frac{104}{3}r^3 = 117$$

$$r^3 = \frac{117 \times 3}{104}$$

$$r^3 = \frac{27}{8}$$

$$r = \frac{3}{2}$$

FINAL ANSWER

$\boxed{1.5 \text{ cm}}$

PAST PAPER QUESTION

Q#: 24 Marks: 5

Topic: **Volume & Surface Area**

Jun 2023 P2HR - Jun 2023 | P2HR

24 The surface area of sphere **A** is nine times the surface area of sphere **B**
The difference between the volume of sphere **A** and the volume of sphere **B** is 117π cm^3

Find the radius of the smaller sphere.
Show your working clearly.

.... cm

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION Q#: 24 Marks: 5

TOPIC: **Volume & Surface Area**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

Let the radius of the smaller sphere be r.
Let the radius of the larger sphere be R.
Surface area of a sphere is

$$4\pi r^2$$

The surface area of sphere A is nine times the surface area of sphere B, so

$$4\pi R^2 = 9(4\pi r^2)$$

$$R^2 = 9r^2$$

$$R = 3r$$

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Cancel π:

$$\frac{104}{3}r^3 = 117$$

$$r^3 = \frac{117 \times 3}{104}$$

$$r^3 = \frac{27}{8}$$

$$r = \frac{3}{2}$$

FINAL ANSWER

$\boxed{1.5 \text{ cm}}$

PAST PAPER QUESTION

Q#: 25 Marks: 6

Topic: **Coordinate Geometry**

Jun 2023 P2HR - Jun 2023 | P2HR

25 The straight line with equation $y - 2x = 7$ is the perpendicular bisector of the line AB where A is the point with coordinates $(j, 7)$ and B is the point with coordinates $(6, k)$

Find the coordinates of the midpoint of the line AB
Show clear algebraic working.

(.................... , ...)

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 6

TOPIC: **Coordinate Geometry**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved to Coordinate Geometry.

Method

The perpendicular bisector is

$$y - 2x = 7$$

so

$$y = 2x + 7$$

The gradient of the perpendicular bisector is 2, so the gradient of AB is

$$-\frac{1}{2}$$

Since $A = (j, 7)$ and $B = (6, k)$,

$$\frac{k-7}{6-j} = -\frac{1}{2}$$

$$2(k-7) = -(6-j)$$

$$j = 2k - 8$$

The midpoint of AB is

$$\left(\frac{j+6}{2}, \frac{7+k}{2}\right)$$

Substitute $j = 2k - 8$:

$$\left(k-1, \frac{k+7}{2}\right)$$

The midpoint lies on $y = 2x + 7$, so

$$\frac{k+7}{2} = 2(k-1) + 7$$

$$k + 7 = 4k + 10$$

$$k = -1$$

Then

$$j = 2(-1) - 8 = -10$$

Therefore the midpoint is

$$\left(\frac{-10+6}{2}, \frac{7+(-1)}{2}\right) = (-2, 3)$$

FINAL ANSWER

$\boxed{(-2, 3)}$

PAST PAPER QUESTION

Q#: 25 Marks: 6

TOPIC: **Coordinate Geometry**

Jun 2023 P2HR - Jun 2023 | P2HR

25 The straight line with equation $y - 2x = 7$ is the perpendicular bisector of the line AB where A is the point with coordinates $(j, 7)$ and B is the point with coordinates $(6, k)$

Find the coordinates of the midpoint of the line AB
Show clear algebraic working.

(.................... , ...)

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 6

TOPIC: **Coordinate Geometry**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved to Coordinate Geometry.

Method

The perpendicular bisector is

$$y - 2x = 7$$

so

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Substitute $j = 2k - 8$:

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Then

$$j = 2(-1) - 8 = -10$$

Therefore the midpoint is

$$\left(\frac{-10+6}{2}, \frac{7+(-1)}{2}\right) = (-2, 3)$$

FINAL ANSWER

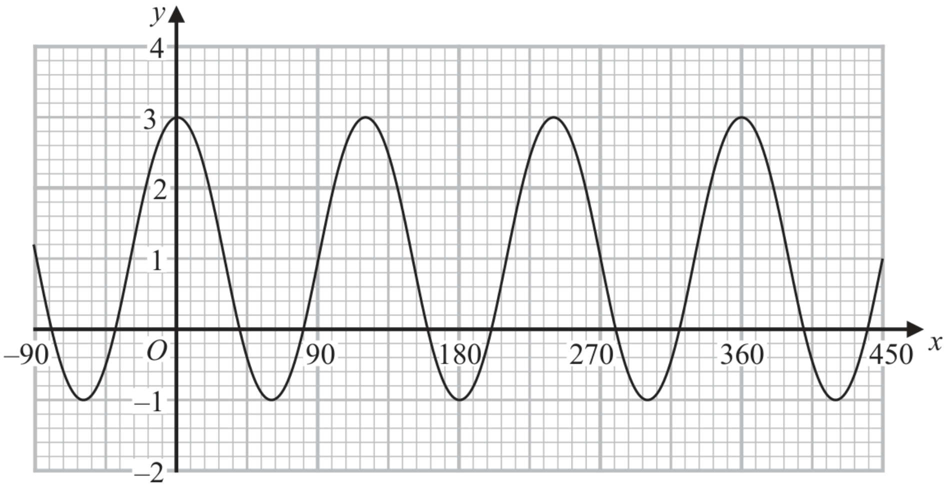
$\boxed{(-2, 3)}$

PAST PAPER QUESTION Q#: 26 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2023 P2HR - Jun 2023 | P2HR

26 Here is a sketch of the curve with equation $y = a\cos bx° + c$ where $-90 \leqslant x \leqslant 450$



Find the value of a, the value of b and the value of c

a = ...

b = ...

c = ...

(Total for Question 26 is 3 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Sine/Cosine Rule to Graphs of Functions.

Method

The curve is

$$y = a\cos bx^\circ + c$$

From the graph, the maximum value is

$$3$$

and the minimum value is

$$-1$$

The amplitude is half the difference:

$$a = \frac{3-(-1)}{2} = 2$$

The midline is the average:

$$c = \frac{3+(-1)}{2} = 1$$

The distance between consecutive maxima is 120°, so the period is

$$120^\circ$$

For $y = a\cos bx^\circ + c$, the period is

$$\frac{360^\circ}{b}$$

So

$$\frac{360}{b} = 120$$

$$b = 3$$

FINAL ANSWER

$\boxed{a = 2,\ b = 3,\ c = 1}$

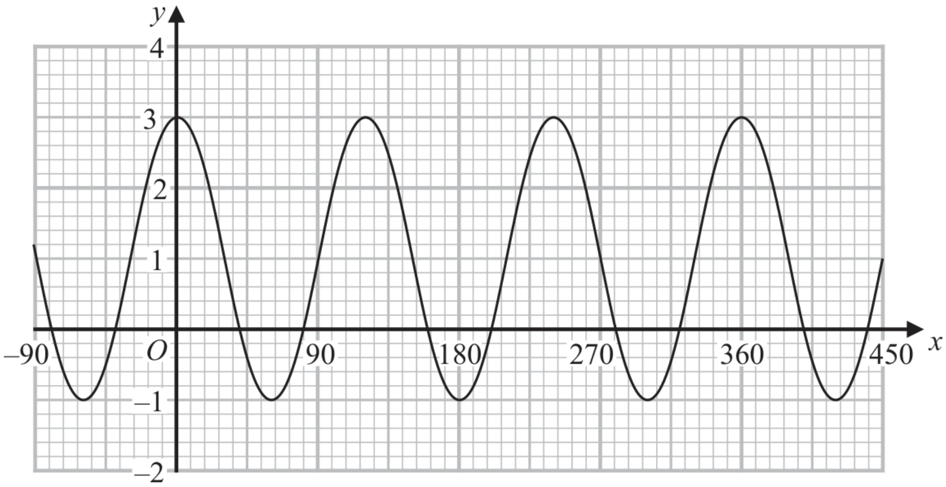
PAST PAPER QUESTION

Q#: 26 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2023 P2HR - Jun 2023 | P2HR

26 Here is a sketch of the curve with equation $y = a\cos bx° + c$ where $-90 \leqslant x \leqslant 450$



Find the value of a, the value of b and the value of c

a = ...

b = ...

c = ...

(Total for Question 26 is 3 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2023 P2HR - Jun 2023 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Sine/Cosine Rule to Graphs of Functions.

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So

$$\frac{360}{b} = 120$$

$$b = 3$$

FINAL ANSWER

$$\boxed{a = 2,\ b = 3,\ c = 1}$$